A Space Transformation-Based Multiform Approach for Multiobjective Feature Selection in High-Dimensional Classification

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***Abstract*—Improving classification performance and reducing the number of selected features are two conflicting objectives of feature selection, which can be well solved by multiobjective algorithms. However, as the dimensionality of the data increases, the search space for feature selection will grow exponentially, which leads to high-computational costs. Additionally, the complex interaction among features makes the population prone to falling into local optimal. To address these issues, feature grouping can treat one dimension as a group of features instead of one feature, effectively transforming the high-dimensional search space into a lower-dimensional one. Since different grouping forms can be converted into different feature combination spaces, the search directions of the population also vary. Inspired by this, a multiform optimization approach based on space transformation (MOFS-MST) is proposed in this article. Specifically, two different grouping forms are set based on the ranking of features in different evaluation criteria to construct a multiform framework, thereby increasing the diversity of the population. During the evolutionary process, a knowledge transfer strategy based on feature groups is executed between the two forms of grouping in order to help each other escape local optima. Moreover, it can dynamically adjust the state of feature grouping to enhance the potential for feature interaction. Experimental results demonstrate that this method outperforms six other state-of-the-art multiobjective high-dimensional feature selection methods on 12 high-dimensional datasets.**

***Index Terms*—Feature selection, high-dimensional classification, multiform optimization, multiobjective optimization.**

I. Introduction

Classification is an important task in the fields of machine learning and data mining, with applications in various domains, including medicine [1], [2], text classification [3], [4] and biology [5], [6]. However, with the development of technology, the emergence of high-dimensional data has become a common phenomenon. These datasets contain many redundant and irrelevant features, which will affect classification performance and increase computational resources [7]. As a preprocessing step in classification, the aim of feature selection is to select a subset of features that are highly relevant and to efficiently remove redundant and irrelevant features, which can improve classification accuracy and train better-classification models [8].

Based on the evaluation criteria, feature selection methods can be broadly categorized into three types: 1) filter-based methods; 2) wrapper-based methods; and 3) embedded-based methods [9]. Filter-based methods evaluate features by various evaluation criteria (such as distance, information gain, etc.) and select features with high scores, so the calculation efficiency is relatively high [10]. The wrapper-based methods find feature subsets through various search methods and train the model for evaluation (such as k-nearest neighbor (KNN), SVM, etc.). Compared with the filter-based methods, they can achieve better-classification performance, but each search needs to build a training model, which leads to higher-computational costs [11]. Embedded-based methods simultaneously balance classification performance and computational efficiency, performing model training and feature selection at the same time [12]. Based on the above analysis, the feature selection problem studied in this article is wrapper-based method.

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In recent years, inspired by biological evolution, evolutionary algorithms (EAs) have been developed rapidly. With the advantages of strong global search ability, no need for domain knowledge and fast search speed, EAs are widely used to solve various complex optimization problems [13], [14], [15]. Since feature selection has two conflicting goals of improving classification accuracy and reducing the number of selected features, it can be regarded as a multiobjective optimization problem [16], [17]. Han et al. [18] proposed a novel multiobjective particle swarm optimization algorithm based on adaptive strategy. First, the selection pressure of the archive is enhanced by an adaptive punishment mechanism based on penalty boundary interaction parameters. Then, the selection pressure of the particles is increased based on the adaptive guidance of the feature information. However, for the high-dimensional feature selection problem, the search space is undoubtedly huge [19], which also brings a series of problems, such as search difficulty and high cost of computing resources. Moreover, in the face of complex interactive features [20], the search direction of the population tends to be simple in the evolutionary process, which will lead to the population is very easy to fall into the local optimal and the diversity of the obtained feature subset is poor. Therefore, how to effectively select a set of feature subsets with high-classification accuracy and small number of features for multiobjective feature selection is difficult in high-dimensional data.

Grouping variables according to certain prior knowledge to reduce the search space is an effective paradigm [21], [22], [23]. For example, Liu et al. [24] overcame the dimensional curse problem by detecting the relationship between decision variables and other variables to group and optimize them in a way of co-evolution. Liu et al. [25] proposed a random dynamic grouping weighted optimization framework to reduce the size of the search space and used the multiobjective particle swarm optimization algorithm for optimization. However, for the new search space formed after grouping, the population may find the local optimal solution, which will have higher requirements for grouping method or space processing method. In addition, the single form transformation also reduces the diversity of feature subsets.

Multiform optimization as a new paradigm, helps algorithms escape local optimal by using different forms of search experience [26], [27], [28]. Moreover, multiform optimization can construct different processing methods, and enhance the diversity of solutions by complementing each other's advantages from different perspectives. Different from multitask optimization, the search space for multiform optimization can be defined by the possible forms of variables and objective functions. For example, Jiao et al. [29] proposed a multiform optimization framework to solve the constrained multiobjective problem by constructing two different tasks to enable the population to search in different sizes of spaces, and using the knowledge transfer of feasible space and infeasible space to find a better solution. Liang et al. [30] proposed a multiform optimization framework to solve the multiobjective feature selection problem. First, a multitask environment with multiobjective feature selection combined with several auxiliary single-objective feature selection is designed to realize knowledge sharing through knowledge transfer. Second, a diversity enhancement mechanism is proposed to find more diverse feature subsets by local searching in promising regions. However, these methods still need to consume a lot of computing resources in the face of huge search space.

Based on the above analysis, a novel multiform approach based on space transformation is proposed in this article. The search space is effectively reduced through feature grouping and different grouping forms are constructed into a multiform framework, thus increasing the diversity of space information. In addition, a novel grouping based knowledge transfer mechanism is proposed to help the population effectively escape from local optimality and enhance the potential of feature interaction during the search process. The main contributions of this article are summarized as follows.

1) A multiform approach based on space transformation is proposed to address the issues of feature selection in high-dimensional classification. Two search spaces with different combinations of features are constructed so that the population can search in multiple directions. Furthermore, by transferring knowledge between the two forms of space translation, they help each other to jump out of the local optimum.
2) Two space transformation strategies are proposed to reduce the search space and enhance the utilization of space information. Features are grouped into different forms based on different evaluation metrics, and one dimension represents a group of features. This process effectively converts a high-dimensional search space into two distinct low-dimensional search spaces. Furthermore, the multiform framework constructed by these two grouping forms helps to enrich the utilization of space information, thereby increasing the diversity of search directions and the possibility of more feature combinations.
3) A novel knowledge transfer mechanism is designed to facilitate information sharing between two distinct grouping forms. This strategy transfers important feature groups between two forms of grouping, helping each other to escape local optima. Additionally, knowledge transfer of feature groups enables dynamic adjustment between groups, thereby enhancing the potential for interaction between features.

II. Background

A. Multiobjective Feature Selection Optimization

Multiobjective EA is widely used in optimization tasks, because it can optimize N objectives simultaneously. Its specific mathematical expression is as follows:

$$\begin{aligned} &\text{minimize} \;\; F(X) = (f_1(X), f_2(X), \ldots, f_N(X)) \\ &\text{subject to} \;\; X = (x_1, x_2, \ldots, x_D) \in \Omega \end{aligned} \tag{1}$$

where D represents the dimension of a solution X in the decision space Ω. For two solutions X_a and X_b, X_a can be considered to dominate X_b if and only if the following conditions are met:

$$\begin{aligned} &\forall i : f_i(X_a) \leq f_i(X_b) \\ &\exists j : f_j(X_a) < f_j(X_b), i, j \in (1, 2, \ldots, N) \end{aligned} \tag{2}$$

In feature selection, $x_D \in \{0, 1\}$, when $x_D = 1$ indicates that the Dth feature is selected, and $x_D = 0$ indicates that the feature is not selected. In addition, for feature selection problems, classification error rate $f_{\text{error}}(X)$ and number of selected features $f_{\text{ratio}}(X)$ are often regarded as two objectives, and the specific expression is as follows:

$$\begin{cases} f_{\text{error}}(X) = \dfrac{N_{\text{err}}}{N_{\text{all}}} \\ \\ f_{\text{ratio}}(X) = \dfrac{\sum_{d=1}^{D} x_d}{D} \end{cases} \tag{3}$$

where N_{err} represents the number of misclassified samples and N_{all} is the number of all samples.

B. Related Study on Multiobjective Feature Selection in High-Dimensional Data

Since feature selection has two contradictory objectives, more and more researchers study multiobjective feature selection. And most algorithms are more biased toward classification accuracy than the number of selected features. However, as high-dimensional data continues to emerge, existing multiobjective approaches not only need to consume a lot of computing resources but also get poor feature subsets in the face of a vast search space. Therefore, in recent years, many excellent algorithms have emerged to solve the problem of high-dimensional feature selection.

There are a lot of redundant features and uncorrelated features in high-dimensional datasets, which seriously affect the classification accuracy. In addition, the number of selected features is one of the objectives of multiobjective feature selection. Therefore, using the principle of sparsity to limit the number of solutions is a good method. For example, Xue et al. [31] proposed a multiobjective feature selection algorithm based on repeated solution analysis. This method restrict the number of selected features in the initialization process to ensure the sparsity of the solution, then filter out redundant solutions through repeatability analysis, and finally propose a diversity protection strategy. However, this initialization method may result in the loss of significant features, and the vast search space increases the likelihood of falling into local optimal. Tian et al. [32] proposed a sparse multiobjective optimization algorithm. It ensures the sparsity of the solution by evaluating the correlation of features and selecting the highly correlated features. Then, the correlation information obtained guides the generation of children. However, for high-dimensional datasets, it takes a lot of computing resources to evaluate each feature. In a word, although the sparsity principle assures the convergence of the number of selected features, it is easy to miss some important features with strong interaction.

The huge search space brings the search difficulty to the high-dimensional feature selection, so the rational reduction of the search space is an effective means to address the high-dimensional problem. To solve this problem, a multiobjective feature selection approach with variable granularity is proposed [23]. The algorithm can significantly reduce the search space by grouping features and representing a group of features with one bit. In the course of evolution, the continuous refinement of the group effectively eliminates unnecessary features that are confused within the group. However, the search space after grouping is relatively simple and the dynamic adjustment of features between groups is lacking, so important features in some groups are easy to be lost. Zhou et al. [33] proposed a discrete feature selection based on multiobjective evolutionary framework. The algorithm maps the search of feature domain to the cut-point domain to reduce the search space, and a flexible cut-point particle swarm algorithm is designed. Finally, a Pareto ensemble method is designed to obtain a better-feature subset. However, for some classes unbalanced data can not get a good cut-point, thus affecting the final performance. Tian et al. [34] proposed a multiobjective algorithm for solving large-scale sparse problems. By mining the maximum and minimum candidate sets of nonzero variables on the Pareto front, and using them to limit the nonzero variable dimension of the solution in the offspring, a binary genetic strategy is designed to improve the quality of the solution. Although the algorithm reduces the search space by using the sparsity of the solution, the complex interaction characteristics of feature selection are not considered. In summary, feature grouping is an effective way to reduce the search space and rapidly locate valuable areas, but the reduced search space is not optimal, so important features are easily lost and feature subsets diversity is poor when the grouping form is unchanged or single.

When many multiobjective feature selection algorithms are applied to high-dimensional problems, the quality of feature subsets obtained is poor because they cannot search the whole space completely [18], [20], [35]. Therefore, better solutions can be found by using prior knowledge or improved guidance strategy. Jiao et al. [36] proposed a multiformal framework to solve the feature selection problem. The framework uses the single objective population to assist the feature selection of the multiobjective population, and then makes use of the advantages of the single objective and the multiobjective population to adjust each other. Finally, the repeated solutions are filtered to improve population diversity. However, this method does not reduce the search space, so it still faces the problem of difficult search for high-dimensional data. Cheng et al. [37] proposed a multiobjective EA that utilizes a steering matrix. The steering matrix of the algorithm is generated according to the importance of each individual and feature. Then the evolution of the population is guided by the steering matrix and the matrix is constantly updated. But the importance of features in the steering matrix is assessed independently, so it is easy to guide populations into local optimal. In conclusion, although the improved guidance strategy can help the population to find a better solution, it still consumes a lot of computational resources and tends to fall into a local optimal when dealing with a large search space.

C. Motivation

The above description reveals that existing methods for high-dimensional feature selection still face challenges due to the vast of the search space. While feature grouping can effectively reduce the search space, the insufficient utilization of space information leads to a tendency toward singular search direction and susceptibility to local optima. To address these limitations, this article proposes a multiform approach based on space transformation. This method utilizes feature grouping to transform high-dimensional search spaces into low-dimensional ones, thereby solving the issues related to difficult search and high-computing resources. Moreover, to overcome the limitations of inadequate space information utilization after grouping, this article introduces the use of various grouping forms to construct multiple search spaces, thus enriching the utilization of space information and diversifying search directions. On this basis, a knowledge transfer strategy based on grouping is proposed. By monitoring the convergence of the population, knowledge transfer between two search spaces is achieved, assisting the population in escaping local optima. In summary, this method effectively reduces the search space while enriching the utilization of space information, thereby enhancing the search efficiency of the population and aiding in effectively escaping local optima.

III. Proposed Approach

A. Framework of the Proposed Approach

Algorithm 1 presents the general framework of the proposed MOFS-MST. The algorithm can be divided into three parts: 1) multiform optimization based on space transformation; 2) knowledge transfer; and 3) detailed search that maps features back to the original search space. In the first part, first, the correlation of features is evaluated and ranked according to symmetric uncertainty (*SU*) [38] and *ReliefF* [39]. Then, G_1 is generated by the proposed nondominated grouping method as the first space transformation form, and G_2 is generated by *K-means* clustering method as the second space transformation form. Finally, the two space transformation forms are constructed into a multiform framework, and one coding bit of an individual represents a group of features, so as to reduce the search space and enrich the utilization of space knowledge (lines 1–5).

In the second part, the Pareto front of a population is monitored in the process of evolution. When the centroid of the Pareto front of the adjacent two generations does not change the distance, it is considered that the population falls into a local optimal and enters the knowledge transfer process. If Pop_1 is considered to be trapped in local optimal, the second grouping form G_2 transfers knowledge to the first grouping form G_1, thereby changing the selection situation of Pop_1 and helping the population escape from the local optimum. The same is true when Pop_2 falls into local optimal (lines 6–22). In addition, population selection will change after knowledge transfer, and features represented by individual genes will also change, so the Pareto front before knowledge transfer will be archived. In the third part, when one of the populations falls into the local optimal β times, the current group can not find a better solution, and more detailed search is needed. Therefore, the features are reverted to the original search space, and the grouping forms of the two populations are eliminated and merged. However, when the distance of the centroid moves more than $\lg D$, the cumulative times are recalculated. The final algorithm outputs the nondominated feature subsets.

Algorithm 1 MOFS-MST

Input: N: population size; $T_{\max}$: maximum number of iterations; *Datesets*: training set.
Output: *DS*: Non-dominated feature subsets.
1: $D \leftarrow$ Dimensions of the datasets;
2: $num_1 \leftarrow 0$; $num_1 \leftarrow 0$;
3: *RK* and $W \leftarrow$ Ranking and normalized evaluation values of all features under the *SU* and *ReliefF* evaluation indicators;
4: $(G_1, G_2) \leftarrow$ *MultiformGrouping* (RK);
5: $(Pop_1, Pop_2) \leftarrow$ *Initialization* (G_1, G_2, N, W);
6: **for** $i = 1$ *to* $T_{\max}$ **do**
7: Updating Pop_1 and Pop_2 through genetic manipulation and environmental selection;
8: Dan_1 and $Dan_2 \leftarrow$ Calculate the distance between the centroid of the current Pareto front and the centroid of the previous Pareto front;
9: **if** $Dan_1 == 0$ **then**
10: $num_1 \leftarrow num_1 + 1$;
11: $A \leftarrow$ Archive operation;
12: $(G_1, Pop_1) \leftarrow$ *Knowledge_Transfer* (G_1, G_2, Pop_1);
13: Updating Pop_1 and Pop_2 through genetic manipulation and environmental selection;
14: **else if** $Dan_1 > \lg D$ **then**
15: $num_1 \leftarrow 0$;
16: **end if**
17: **if** $Dan_2 == 0$ **then**
18: $num_2 \leftarrow num_2 + 1$;
19: $A \leftarrow$ Archive operation;
20: $(G_2, Pop_2) \leftarrow$ *Knowledge_Transfer* (G_1, G_2, Pop_2);
21: Updating Pop_1 and Pop_2 through genetic manipulation and environmental selection;
22: **else if** $Dan_2 > \lg D$ **then**
23: $num_2 \leftarrow 0$;
24: **end if**
25: **if** $num_1 > \beta$ or $num_2 > \beta$ **then**
26: $P_1, P_2 \leftarrow$ Remove the grouping and map Pop_1 and Pop_2 back to the original space;
27: $P \leftarrow P_1 \cup P_2$;
28: Updating P through genetic manipulation and environmental selection;
29: **end if**
30: **end for**
31: $DS \leftarrow$ Obtain the non-dominated feature subsets;
32: **return** *DS*;

B. Multiform Optimization Based on Space Transformation

The core concept of multiform optimization lies in addressing a specific optimization problem not only within the confines of a single problem formulation or model, but rather by exploring multiple different representations of the problem through extracting useful knowledge from the original task. Moreover, it involves simultaneously optimizing these alternative representations to improve the performance of the entire problem solving process. Furthermore, compared to multitask optimization, multiform optimization focuses more on multiple representations of a single problem rather than concurrently solving multiple distinct optimization problems.

For the high-dimensional feature selection problem, the search space can be reduced effectively by using feature grouping to treat each group of features as one dimension. However, the single grouping method makes the transformed

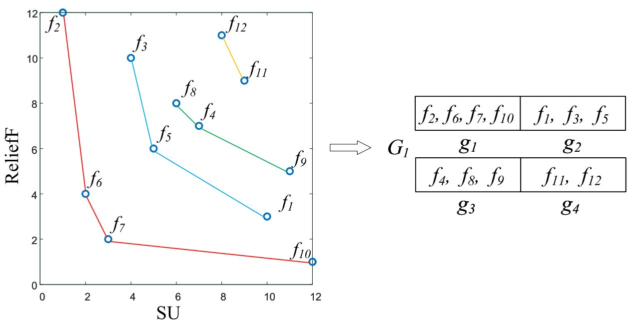

Fig. 1. Features are grouped based on nondominated sort. The horizontal and vertical coordinates represent the *SU* and *ReliefF* rankings, respectively. Different colors indicate different groups, G_1 indicates the first group form, there are four feature groups g_1, g_2, g_3, and g_4.

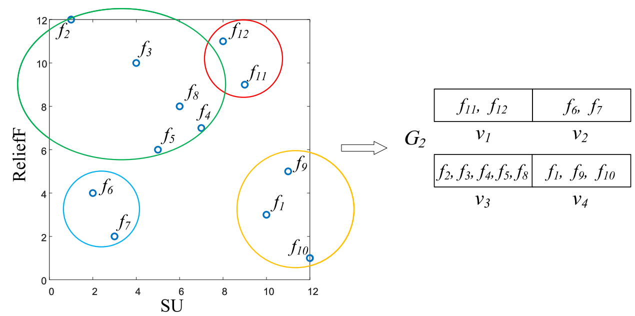

Fig. 2. Features are grouped based on *K-means* forms. G_2 indicates the second grouping form, with four feature groups v_1, v_2, v_3, and v_4.

search space relatively fixed, and the solutions that may be found are not good and lack diversity. Multiform approach can effectively address these issues. Therefore, a multiform optimization approach based on space transformation is designed to get better-combined feature subsets.

At the beginning, the correlation rankings of *SU* and *ReliefF* for each feature are used as the basis for two forms of grouping. The choice of *SU* and *ReliefF* is motivated by two reasons.

1) *SU* measures the importance of features by capturing their nonlinear relationship with the label, while *ReliefF* evaluates feature importance by the ability of the features to distinguish nearby samples. Considering these different perspectives allow for a comprehensive analysis of the problem.
2) *SU* and *ReliefF* are widely employed for feature selection due to their proven effectiveness. Subsequently, the rankings obtained from *SU* and *ReliefF* are used as horizontal and vertical axes, respectively.

And all features are mapped to a two-dimensional (2-D) coordinate system according to the two rankings.

1) *Define the First Form of Feature Grouping:* Since the evaluation angles of *SU* and *ReliefF* are different, the rankings of these two evaluation indicators should be weighed at the same time when grouping, so a novel grouping method is designed. This method employs the concept of nondominated sorting by treating each feature as a solution and the *SU* and *ReliefF* evaluation rankings of each feature as two objective values. Subsequently, all features are subjected to nondominated sorting and divided into different layers. The features on the Pareto front represent the most valuable layer after weighing the *SU* and *ReliefF* rankings, while the second layer is considered to be inferior, and the features in the last layer are deemed to have the lowest value. Finally, each layer is treated as one feature group, thereby completing the division of all features. At the same time, nondominated sorting can automatically determine the number of layers, and each layer has a different degree of importance, so the grouping number Gnum can be adaptively determined. Additionally, it simultaneously considers two different evaluation methods.

Fig. 1 shows the first grouping form G_1, in which 12 features are grouped based on the idea of nondominant sort, and the features are adaptively divided into four layers. The features of each layer are treated as a group, thus dividing the features into $\{g_1, g_2, g_3, g_4\}$.

2) *Define the Second Form of Feature Grouping:* Since *K-means* is a convenient and widely used clustering method, it is adopted as the second form of grouping. Specifically, the number of groups Gnum can be obtained adaptively from the first grouping form. As a result, K is set to Gnum. Subsequently, *K-means* clustering is utilized to divide the features into Gnum groups adaptively. It is worth noting that each grouping form is assigned a separate population to search, and the size of these two populations, denoted as N, is equal.

Fig. 2 represents the second grouping form G_2, where 12 features are grouped into four groups based on *K-means* clustering, et. $\{v_1, v_2, v_3, v_4\}$. The size of the search space is converted from 2^{12} to 2^4.

3) *Multiform Optimization:* A multiform optimization approach is constructed based on the above two space transformation forms. By grouping D features into Gnum groups and treating each group as one dimension, the search space is reduced from 2^D to 2^{Gnum}. However, the search space formed by each grouping form has its own characteristics. Specifically:

1) The first grouping form is a layered division of features through nondominated ideas, where features within groups are widely distributed on the 2-D axes of *SU* and *ReliefF*. The second grouping method is based on the *K-means* approach, resulting in a more concentrated distribution of features. Therefore, the two grouping methods have different perspectives;
2) The first grouping method yields groups of features with similar importance after integrating two evaluation metrics. The second method produces groups of features with similar rankings but unequal importance;
3) The features contained in the groups of the two forms are different, and individuals with the same code can obtain different feature combinations in different spaces, enriching the diversity of feature combinations.

Considering the above analysis, both forms of grouping have unique advantages and can complement each other during the evolutionary process. Therefore, they are combined into a multiform optimization framework. Additionally, populations can obtain different information in different search spaces, enriching the diversity of search directions and feature combinations, thus improving search efficiency.

Then, the search of populations in this approach is optimized. It is remarkable that the two populations evolved independently. As each group of features is considered as one dimension, it is essential to take into account the characteristics of all features within a group during the search process. When an individual is initialized, the threshold for each group is set as the average value derived from the *SU* and *ReliefF* evaluation of all the features within that group. Then generate a random number of 0 to 1, and if the random number is greater than the obtained threshold, this group are selected, that is, this group code is 1, otherwise not selected. Thus, during initialization, a large number of irrelevant features are eliminated. In addition, the algorithm adopts NSGA-II as the basic optimization algorithm and makes improvement on this basis. First, the proposed algorithm uses binary encoding and employs binary tournament to select parents. Then, each bit of the parents generates two offspring. If both parents select a certain group of features, it is proved that the group is more important and the offspring still select it. On the contrary, if none of the parents select a certain group of features, it is considered irrelevant or redundant and the offspring do not make a selection. If the selections are different, the selections of the offspring are determined by the probability P_c. Finally, the mutation probability P_m is set to $1/\text{Gnum}$. The setting of P_c is as follows:

$$P_c = \frac{\text{avgRK}_1^j + \text{avgRK}_2^j}{2 * D} \tag{4}$$

$$\text{avgRK}_1^j = \frac{\sum_{n=1}^{k} R_1^j(n)}{k} \tag{5}$$

$$\text{avgRK}_2^j = \frac{\sum_{n=1}^{k} R_2^j(n)}{k} \tag{6}$$

where avgRK_1^j and avgRK_2^j represent the average value of all features in jth ($j = 1, 2, \ldots, \text{Gnum}$) group based on *SU* and *ReliefF* ranking, respectively. k represents the number of features in jth group. R_1^j and R_2^j represent the ranking of the feature based on *SU* and *ReliefF* in the jth group, respectively. It can be seen from (4) that a smaller P_c value represents a higher ranking of features in jth group. In addition, when the random number *rand* is greater than P_c, the group is selected, so that important feature groups will be retained, while the feature groups with low correlation are more likely to be eliminated. At the same time, the two proposed grouping methods divide unimportant features and important features, and the population can obtain a better-Pareto frontier through iteration, so a large number of irrelevant and redundant features are quickly eliminated.

C. Knowledge Transfer

Due to the reduction of search space, populations corresponding to different grouping forms can quickly locate valuable feature groups. However, populations are still easy to fall into local optimal during their respective search processes. Additionally, each dimension represents a group of features, where the dimension is assigned a value of either 1 or 0, indicating the selection or exclusion of all features within that group. It cannot be guaranteed that the selected groups do not contain redundant or unrelated features, or that the unselected groups do not contain strongly interacting features. This results in the loss of some information. Therefore, it is necessary to adjust the grouping so that important features can be in the same group with a high probability. To this end, a novel group-based knowledge transfer strategy is designed to facilitate populations learn from each other to get out of the local optimal. Simultaneously, the dynamic adjustment of the feature groups is achieved in the process of knowledge transfer.

Knowledge transfer involves three problems that need to be considered. First, judging when knowledge should be transferred. Inappropriate timing of transfer can impact the population's search direction. Second, what knowledge can be transferred between two grouping forms. Since not all knowledge contributes to the population's evolution, careful selection of useful knowledge can avoid the impact of negative transfer. Third, how the selected useful knowledge can be transferred between two grouping forms. The above issues will be described in detail in the following.

1) *When to Transfer:* Since the population easily fall into local optimal, it is appropriate to trigger the knowledge transfer strategy at this time. Therefore, this study examines the differences between the current Pareto front and the previous generation's Pareto front to determine whether the population is in a local optimum. Specifically, the average of the horizontal and vertical coordinates of the solutions on each generation of Pareto frontier is calculated, and this average is used as the centroid of the frontier. If the Euclidean distance between the centroids of two adjacent generations is 0, it indicates that the population has not discovered a better solution. In this case, it is considered to be in a local optimum, necessitating the learning from another population.

2) *What to Transfer:* Determining what knowledge helps another population learn is critical. Moreover, due to different grouping forms, individual genes in different populations represent different feature groups, so it is unreasonable to directly transfer individual genes or complete individuals. Furthermore, since the selection of all features within a group remains consistent, it is desired to have important and unimportant features allocated to different groups. Based on the above discussion, this study focuses on transferring the significant feature groups. Specifically, in the current generation, the number of times each feature group is selected is calculated. Then the feature groups of the source task are sorted in descending order based on the number of selected times. Among them, the top feature groups are considered valuable and the feature division of these groups is considered reasonable. Therefore, these feature groups will be transferred as valuable knowledge. The number of groups to be transferred is set to α. Algorithm 2 provides an example of knowledge transfer from G_2 to G_1, with the process elaborated in lines 1–3. Besides, if there are multiple groups with a ranking of α, the group with the fewest features is chosen to minimize the possibility of unwanted features being included. It is worth noting that if the selected feature group has already been transferred, the subsequent group in the rank will be chosen, and so on.

Algorithm 2 Knowledge Transfer

Input: G_1; G_2; Pop_1;
Output: $NewG_1$; $NewPop_1$;
1: $sum\_select \leftarrow$ Count how many individuals are selected for each group in G_2;
2: $S\_G_2 \leftarrow sort(sum\_select)$; // Sort each group by the selected number in descending order.
3: $Trans\_g \leftarrow S\_G_2\{1:\alpha\}$; // The top α groups are selected as the transferred groups.
4: **for** $i = 1$ *to* $numel(Trans\_g)$ **do**
5: $B \leftarrow Trans\_g\{i\}$; // Features of group i in the transferred group.
6: $NewG_1 \leftarrow$ Features in set B are picked from G_1 and new groups are formed;
7: **end for**
8: $L \leftarrow numel(G_1)$;
9: $incr\_L \leftarrow numel(NewG_1) - numel(G_1)$; // Number of new groups added to G_1 after transfer.
10: $NNF \leftarrow$ Find individuals outside the non-dominant front of Pop_1.
11: **for** $i = 1$ *to* $incr\_L$ **do**
12: $Pop_1(NNF, L+i) \leftarrow 1$; // The transferred groups are selected by individuals in NNF.
13: $NewPop_1 \leftarrow Pop_1$;
14: **end for**
15: **return** $NewG_1$; $NewPop_1$

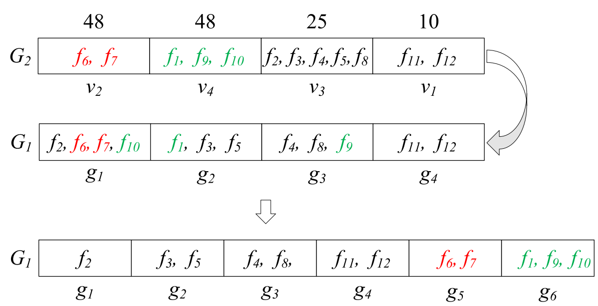

Fig. 3. Knowledge transfer based on multiform grouping. The features in red and green represent the first and second group of features that need to be transferred, respectively.

3) *How to Transfer:* The selection of an appropriate transfer method significantly enhances knowledge acquisition. As the transferred groups are considered to have a reasonable feature division, it is desired to have the same feature division in another grouping form. First, look for which features exist in the transferred groups that are being transferred. Next, select these features from another grouping form to recreate the same α groups. Lastly, for the newly added α groups, individuals on the Pareto front in the population are not selected, while other individuals undergo selection (lines 4–14 of Algorithm 2). This behavior ensures that the population's search direction does not fluctuate significantly. Furthermore, since the newly formed α groups are recombined by extracting features from different groups, this knowledge transfer mechanism can achieve dynamic adjustment between groups, which effectively increases the probability of more feature interactions and better separating out irrelevant features. At the same time, the scores of redundant features are similar according to the evaluation indicators, so there are more redundant features in one group. Therefore, the number of features in some groups is refined, which can eliminate redundant features more effectively.

Fig. 3 provides a simple legend. It is assumed that the population of the first grouping form G_1 falls into a local optimal, so G_2 is required to transfer knowledge to G_1. And assume that the number of groups α to be transferred is two. In the grouping form represented by G_2, the number of selected individuals in each group is sorted. v_2 and v_4 are selected by 48 individuals, so it is considered that the grouping of these two groups is correct. In this regard, the features of v_2 and v_4 are selected from the grouping form represented by G_1, and these features are divided according to the grouping of v_2 and v_4. g_5 and g_6 are newly generated groups. It can be clearly seen from the figure that except for g_4, other groups have dynamic adjustment. Therefore, this knowledge transfer strategy not only uses mutual learning to escape the local optimal, but also achieves the dynamic adjustment of the group. In addition, the number of features in g_1, g_2 and g_3 become smaller, which will contribute to a more detailed exploration of the features within these groups by the population.

As the population continues to evolve toward a better-Pareto front, there will be some groups selected by all and some groups discarded by all. This represents an undesirable extreme scenario. Because the unselected group may possess individually important features, while the selected group may contain redundant features. For this, in the first genetic operation after knowledge transfer, the group selected by all individuals and the group abandoned by all individuals have a flipping probability of $1/\ln D$. This operation increases the probability that some potentially valuable groups will be migrated and the diversity of the population. Furthermore, when the cumulative number of times that the centroid of the Pareto front moves to zero is β, it is considered that the current grouping form cannot find a better solution through knowledge transfer and evolution. In addition, there is no guarantee that the features within a group are all valuable, and if you simply remove one grouping form, the features in the other grouping form will still not be searchable in more detail. Therefore, both grouping forms are eliminated, i.e., one dimension representing one feature, for the final detailed search. If the centroid is moved more than $\lg D$, the cumulative number is recalculated.

D. Analysis of Computational Complexity

For computational complexity, this article will analyze the rest except for the training and evaluation of the classifier. Feature grouping is executed only once during initialization, regardless of its computational complexity. The computational complexity of the genetic operator is $O(\text{Gnum} * N)$, where Gnum represents the number of current feature groups and N represents the sum of the sizes of the two populations. The computational complexity of environment selection in the proposed approach is $O(N^2)$. The computational complexity of knowledge transfer is $O(\text{Gnum})$. Thus, the computational complexity of the proposed MOFS-MST is $O(N^2)$. The computational complexity is the same as that of existing multiobjective algorithms based on NSGA-II.

IV. Experimental Design

A. Datasets

A total of 12 high-dimensional datasets from the real world are used for the experimental analysis. The dimensions of

TABLE I
BASIC INFORMATION OF THE 12 DATASETS

Datasets	#Features	#Instances	#Classes	%Smallest class	%Largest class
GLIOMA	4,434	50	4	14	30
Leukemia1	5,327	72	3	13	53
9Tumor	5,726	60	9	3	15
Brain1	5,920	90	5	4	67
tumors_C	7,129	60	2	35	65
Carcinom	9,182	174	11	3	16
nci9	9,712	60	9	3	15
arcene	10,000	100	2	44	56
Brain2	10,367	50	4	14	30
Leukemia2	11,225	72	3	28	39
11Tumor	12,533	174	11	4	16
GLI_85	22,283	85	2	31	69

these datasets range from 4,434 to 22 283. They are the open resources of http://featureselection.asu.edu and https://www.openml.org/search. Moreover, the 12 datasets are highly unbalanced, thus adding difficulty to the classification process [40]. Details are shown in Table I.

B. Comparative Approaches and Parameter Settings

Six state-of-the-art high-dimensional multiobjective algorithms are used as comparison algorithms in this study. The details are as follows: a steering-matrix-based multiobjective EA (SM-MOEA) [37]. A EA based on duplication analysis (DAEA) [31]. A variable granularity search-based multiobjective feature selection algorithm (VGS-MOEA) [23]. A multiform approach for feature selection (MFFS) [36]. Large-scale sparse multiobjective algorithm based on pattern mining (PMMOEA) [34]. A multiform optimization framework to solve the multiobjective feature selection (MFMOEA) [30].

The reason for choosing the above six algorithms is that VGS-MOEA is newly proposed to address the high-dimensional feature selection issues by using grouping idea, and achieves excellent results. MFFS and MFMOEA are newly proposed multiform method for feature selection problems, which also validates high-dimensional data. The other three algorithms are all newly proposed multiobjective high-dimensional feature selection algorithms, which can make the proposed MOFS-MST more convincing.

The parameters involved in the comparison algorithm follow the original papers. The parameters of MOFS-MST are determined adaptively, except for the number of transferred groups α. Both the population size N and the number of iterations T are set to 100 in this study. Since the algorithm involved two populations, in order to ensure the fairness of the experiment, the size of the two populations is set to 50, respectively, and the total number is still 100. Except that the threshold of the proposed algorithm adopts adaptive determination, the threshold of other algorithms is uniformly set at 0.5. Each data set was randomly divided into a 70% training set and a 30% test set. Among them, only the training set is involved in the evolution and evaluation of the population, and the test set is only tested when the final feature subset is obtained to ensure the universal applicability of the obtained feature subset. In model training and evaluation, 5-fold cross-validation is used to avoid feature selection bias. In this article, the KNN [41] is used as the classifier to calculate the classification accuracy, because the research object of this article is high-dimensional data, and the simple and fast advantage of KNN can improve the efficiency of algorithm verification, where k is set to 5 [42].

Hypervolume (HV) [43] and inverted generational distance (IGD) [44] are used as experimental metrics. Both indicators are used to verify the convergence and diversity of the obtained Pareto front. For HV, the resulting objective values are normalized to a scale of [0, 1] in each objective direction, and the reference point is set to (1, 1). For the calculation of IGD, since the true Pareto front of high-dimensional feature selection is almost impossible to find, according to the suggestion of [45], the Pareto front obtained by running each algorithm 30 times is merged into a joint population, and the nondominated solution obtained after the combination is taken as the true Pareto front.

C. Sensitivity Analysis of Parameters

The number of transferred groups α and β are important parameters in this algorithm. The size of α determines how much content is learned from each other, while the size of β decides when to discontinue the grouped form to conduct a final detailed search. Therefore, sensitivity analysis experiments for these parameters will be performed in this section. The results are the mean HV values for each parameter over 30 runs on the 12 datasets. It is shown in the Table S-I of the MSTsupplementary file, where the bold results represent the best values among the different α and β.

When the source task transfers a group of features, the target task needs to identify and select all the features in the transferred feature group. Since these features are distributed across various groups in the target task, even transferring a small number of groups can dynamically affect the combination and selection of features in each group. For this experiment, α is set as {0, 1, 2, 3} and β is set to {2, 3, 4}. The experimental results indicate that when α is 1 and β is 3, the algorithm performs optimally on 8 out of 12 datasets. When α is set to 0, it means that no transfer of feature groups is performed, and the result has no advantage, proving the effectiveness of feature group transfer. It is known from the proposed strategy that the first grouping form and the second grouping form will have the same α groups after transferring. Therefore, the larger α is, the more the number of the same groups will be after transferring, which will not help the population jump out of the local optimal in the later stage. In addition, the newly formed feature groups need to be reselected after transfer, and if too many groups are transferred there is a greater impact on the population search direction. Based on the above analysis and experimental results, $\alpha = 1$ is a better choice. When β is set to a low value, the population may not fully explore the reduced space, and discontinuing grouping at this stage can increase the difficulty of subsequent searches while consuming computational resources. Conversely, if β is set too high, the population might not have the opportunity to conduct a detailed search within T, or there may be too few generations of search after grouping is discontinued, resulting

TABLE II
AVERAGE HV COMPARISON RESULTS ON THE TEST DATA

Datasets	SMMOEA	DAEA	VGS-MOEA	MFFS	PMMOEA	MFMOEA	MOFS-MST
+/=/-	12/0/0	11/1/0	11/0/1	11/1/0	8/3/1	10/1/1	-
Ranking	6.3333	3.7083	4.9583	5.2083	3.3333	3.125	**1.3333**

TABLE III
AVERAGE IGD COMPARISON RESULTS ON THE TEST DATA

Datesets	SMMOEA	DAEA	VGS-MOEA	MFFS	PMMOEA	MFMOEA	MOFS-MST
+/=/-	10/1/1	10/2/0	11/1/0	10/2/0	7/4/1	11/1/0	-
Ranking	6.375	3.9167	5.1667	4.9583	3.0833	2.75	**1.75**

in insufficient detail in the search. Therefore, setting β to 3 is considered appropriate.

V. EXPERIMENTAL RESULTS AND PERFORMANCE ANALYSIS

This section presents a comparison between the proposed algorithms and six state-of-the-art algorithms mentioned in the previous section. The indicators of comparison are HV, IGD, Pareto front, maximum classification accuracy (MCA), number of selected features and running time of each algorithm on 12 datasets. The results are compared by Wilcoxon's rank-sum test at the significance level of 0.05, and Friedman test [46] is used to rank each algorithm, represented by ranking in the table. In addition, "+", "=," and "−", respectively, indicate that the proposed algorithm is significantly better than, approximately equal to, and significantly worse than the comparison algorithm. The bold font in each table indicates the optimal result. Finally, the effectiveness of the proposed strategies are verified and the reasons for their effectiveness are analyzed.

A. Comparisons in Terms of the Multiobjective Performance Metrics

Table S-II of the MSTsupplementary file presents the average HV of all algorithms after 30 runs on 12 datasets. A higher-HV value indicates better-algorithm diversity and convergence. Table II shows the simple results. It can be seen from Wilcoxon's rank-sum test that the proposed algorithm significantly outperforms the other six comparison algorithms on eight datasets. And Friedman's ranking results also validate MOFS-MST ranks first. Although DAEA performs better than the proposed algorithm in terms of HV on the GLI_85 dataset, there is no significant difference between DAEA and the proposed algorithm. This is because DAEA develops strategies for repetitive solutions to increase the diversity of knowledge, but the large search space still presents it with a difficult search problem. On the Leukemia1 dataset, VGS-MOEA demonstrates superiority over MOFS-MST due to its use of the grouping idea and better sparsity. However, VGS-MOEA uses only one grouping method and does not dynamically adjust the groups, so it is significantly worse than MOFS-MST on the other 11 datasets. MOFS-MST is inferior to MFMOEA on the 11Tumor dataset because MOFS-MST reselected the features to ensure the quality of the feature subset, resulting in a large number of selected features. PMMOEA is an algorithm designed for large-scale multiobjective problem. The number of features selected by this algorithm is very small and considers the quality of the selected features. Therefore, the HV values on tumors_C, Leukemia1 and GLI_85 datasets are similar to MOFS-MST, but still perform poorly, and only on Brain1 performs better. This is because MOFS-MST uses the grouping idea to reduce the search space, and presents a multiform framework to make the search more diverse. At the same time, the knowledge transfer strategy can make the population learn from each other, and realize dynamic adjustment between groups, which will help to improve the possibility of interaction between features. In general, the proposed algorithm has obvious advantages on most datasets, indicating that it is well convergence and diversity.

Table S-III of the MSTsupplementary file lists the average IGD results after 30 runs of all the algorithms. A lower-IGD value indicates a closer approximation to the true Pareto front. It can be seen from Table III that MOFS-MST performs best on eight of the 12 data sets. For the datasets 11Tumor, SMMOEA, and MFMOEA performs well because they use the guidance strategy to mine the better-feature subsets and ensure that the number of selected features is small. However, on high-dimensional datasets, the huge search space still causes these two algorithms to fall into local optimal easily, so the overall performance is not better than MOFS-MST. Because of its high sparsity, PMMOEA performs better on the objective with the number of selected features. However, the feature selection problem leans more toward classification accuracy rather than the quantity of selected features. MOF-MST filters out a large number of features by threshold during initialization to ensure a small number of features. In addition, multiform optimization based on spatce transformation solves the problem of large search space and enriches the utilization of space knowledge. Finally, knowledge transfer can help populations jump out of local optimality to improve classification accuracy. Consequently, most datasets with similar IGD results do not outperform MOFS-MST. In summary, the proposed MOFS-MST approach exhibits strong performance.

B. Nondominated Front Distribution Analysis

Due to space constraints, Fig. 4 selects the Pareto frontiers of four representative datasets. Fig. S-I of the MSTsupplementary file presents the Pareto front plots with the median HV values from all the algorithms after 30 runs on the 12 datasets. It is not difficult to see from the figure that the proposed MOFS-MST algorithm can obtain better-Pareto front solution sets compared with DAEA, VGS-MOEA, MFFS, and MFMOEA on most datasets. Among them, on the datasets GLIOMA, nci9, Brain2, 11Tumor, and GLI_85, MOFS-MST has no obvious advantage, but the Pareto front solutions can obtain higher-classification accuracy and better diversity. PMMOEA and SMMOEA have better-Pareto fronts on some datasets than MOFS-MST. This is because PMMOEA adopts the principle of sparsity. The guiding strategy of SMMOEA enables the population to select fewer features, but the population is easy to fall into local optimality. Therefore, the classification accuracy and diversity are significantly lower

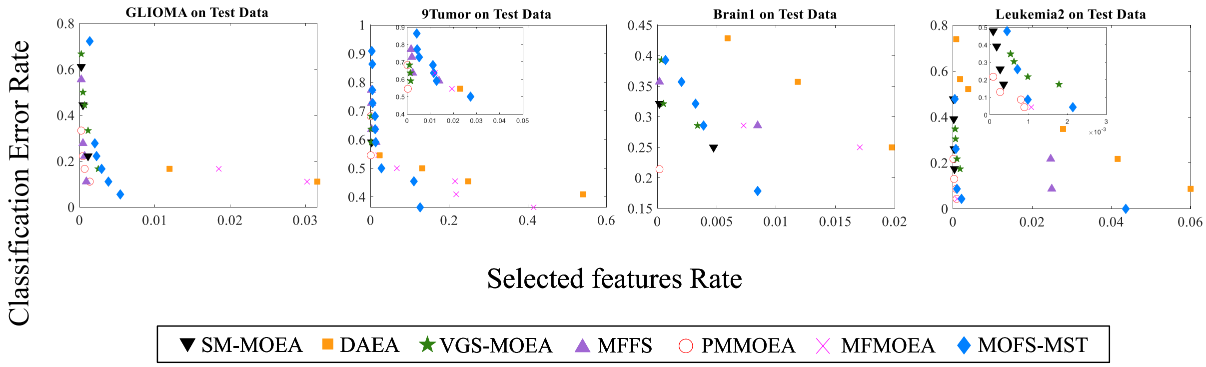

Fig. 4. Distributions of nondominated feature subsets obtained by each approach on test sets in terms of median HV value.

TABLE IV
AVERAGE MCA AND FN OF THE SIX COMPARISON ALGORITHMS ON TEST DATA

Datasets	SMMOEA		DAEA		VGS-MOEA		MFFS		PMMOEA		MFMOEA		MOFS-MST	
	MCA	FN	MCA	FN	MCA	FN	MCA	FN	MCA	FN	MCA	FN	MCA	FN
Average	72.09%	21.58	80.60%	915.08	74.59%	45.75	75.40%	97.67	78.87%	**9.67**	82.43%	549.60	**85.18%**	267.42
Ranking	6.4167	-	3.5	-	5	-	5.3333	-	3.5833	-	2.8333	-	**1.3333**	-

than that of MOFS-MST. The feature subsets obtained in this article tend to improve the classification accuracy rather than the number of selected features. In general, the Pareto front solutions obtained by MOFS-MST have better diversity and classification accuracy, which also verifies that the proposed multiform grouping strategy can provide multiple search spaces, thereby increasing the diversity of solutions. And the proposed knowledge transfer strategy can help the population to get higher-classification accuracy.

C. Classification Performance

Table S-V of the MSTsupplementary file compares the MCA of the feature subset obtained by each algorithm on the test sets and the number of selected features (FN) corresponding to the MCA. Table IV shows the simple results. The results in the table are derived from the average MCA after 30 runs and the corresponding average FN. It can be observed that MOFS-MST has the highest MCA in 8 of the 12 datasets, which has a significant advantage over the other six comparison algorithms. According to Friedman's test results, MOFS-MST ranked first out of seven algorithms. This result proves that MOFS-MST has higher-classification accuracy. MOFS-MST is not dominant in the number of selected features, because the newly generated feature sets are reselected after knowledge transfer. SMMOEA, VGS-MOEA, and PMMOEA are more focused on the principle of sparsity. However, getting poor classification accuracy with a very small number of features is not the desired result. Maximizing classification accuracy is more important and challenging than minimizing the number of selected features [45]. MOFS-MST solves the problem of insufficient utilization of space knowledge and easy to fall into local optimal by means of space transformation and knowledge transfer. Therefore, the proposed algorithm still has good usability.

TABLE V
TRAINING TIME CONSUMED BY THE SEVEN APPROACHES ON THE 12 DATASETS (UNIT: S)

Datasets	SMMOEA	DAEA	VGS-MOEA	MFFS	PMMOEA	MFMOEA	MOFS-MST
Average	437.54	611.59	421.46	599.77	627.19	467.08	**355.23**

D. Computational Times

Computing time is a key index to measure the computing efficiency of an algorithm. Therefore, a comparative experiment is designed to compare the running time of the algorithms. As shown in the Table S-VI of the MSTsupplementary file, the results are obtained from running each algorithm on 12 datasets and averaging the times over 30 runs. Table V shows the simple results. It can be seen from the table that MOFS-MST consumes significantly less time than other algorithms in seven out of the 12 datasets. In addition, the average value of the total running time of each algorithm on all datasets is calculated, and MOFS-MST still takes the shortest time. VGS-MOEA is also a feature selection algorithm by grouping, so it takes the second most time. This also proves that the search space can be effectively reduced by grouping, and the population can quickly locate the valuable search area. However, MOFS-MST preprocesses the threshold value during initialization, thus filtering out a large number of unimportant features, speeding up the convergence of the number of selected features, and reducing the training time of the classifier, so the time consumption is shorter than that of VGS-MOEA. The experiment proves that the proposed MOFS-MST algorithm can find more excellent feature subsets on high-dimensional data sets with less computing time.

E. Effectiveness of the Designed Strategies

In this article, the multiform grouping strategy is used to reduce the search space and increase the diversity of the population. In addition, a knowledge transfer mechanism based

TABLE VI
MEAN HV PERFORMANCE OF MOFS-MST VARIANT AND MOFS-MST COMPARISON RESULTS ON 12 DATASETS

Verification	Algorithm Comparison	HV (+/-/=)
Effectiveness of the Different Grouping Strategies	Kmeans-Group vs ND-Group	3/8/1
	Random-Group vs ND-Group	5/5/2
Effectiveness of the Knowledge Transfer Mechanism	MOFS-NTK vs MOFS-MST	9/3/0

on grouping is proposed to help the population jump out of the local optimal and keep more important features in the same group. This section verifies the validity of the proposed strategy. All results are derived from the average HV values of 30 runs on 12 datasets. The comparison results are shown in Table VI.

1) The Strategy for Different Grouping Methods: In multiform optimization strategy, this article presents a grouping method based on the idea of nondominant sorting, and another grouping method uses *K-means* clustering. To verify the effectiveness of these two grouping strategies, they are compared with Random grouping. Random grouping refers to randomly selecting one of the Gnum groups for each feature and placing it in the group, ensuring that at least one feature exists in each group. As shown in the Table S-IV of the MSTsupplementary file, the three grouping methods are all tested on the framework of NSGA-II [47], and the number of groups is consistent across all grouping methods. Among them, ND-Group is a grouping method based on the idea of nondominated sorting proposed in this article. The experimental results demonstrate that the ND-Group outperforms all other methods in seven out of the 12 datasets, with 5 of them showing significant improvement over the Random-Group. This is because the proposed ND-Group grouping method considers two different evaluation indicators at the same time, and uses the idea of nondomination sorting to divide features of different importance into different levels. In addition, ND-Group can adapt to determine the number of groups, avoiding the influence of artificial setting. As can be seen from Friedman's Ranking, ND-Group ranks first, Kmeans-Group ranks second, and Random-Group performs the worst. Therefore, this article chooses ND-Group and Kmeans-Group to construct a multiform grouping framework.

2) The Mechanism for Knowledge Transfer: Since the proposed knowledge transfer mechanism is completed on the grouping strategy, only the ablation experiment of this mechanism is carried out, as shown in the Table S-VII of the MSTsupplementary file. MOFS-NTK indicates the algorithm after the knowledge transfer mechanism is removed. It is obvious from the table that after adding the knowledge transfer mechanism, MOFS-MST is the best in 11 of the 12 datasets, and 9 of the datasets are significantly better than MOFS-NTK. Among them, the poor performance on dataset Brain1 may be due to the reselection of newly generated groups after transferring, which increases the number of selected features, but there is no significant difference. This result clearly shows that the proposed knowledge transfer mechanism significantly improves the performance of the algorithm. This is because this strategy can effectively help the population to jump out of the local optimal, and the dynamic adjustment of feature groups can help more important features to be divided into the same group and enhance the potential for feature interaction.

VI. CONCLUSION

This article introduces a novel approach to tackle the high-dimensional feature selection problem through a multiform approach based on space transformation. First, multiform optimization based on space transformation is proposed. Based on two evaluation indicators, the features are grouped using the idea of nondominant sorting and the K-means method, thereby defining two different forms of space transformation. And one group of features is regarded as one dimension, effectively transforming a high-dimensional search space into two low-dimensional search spaces. Additionally, these two forms of space transformation have their respective advantages, thus constituting a multiform optimization framework. The population can perform searches in different directions within the different feature combination spaces, thereby increasing the diversity of the population. Finally, a knowledge transfer mechanism based on groupings is proposed, which realizes the dynamic adjustment of feature groups by transferring knowledge to excellent groups. This mechanism can not only help the population jump out of the local optimal, but also make more important features in the same group.

The performance of the proposed algorithm is validated by comparing it with six state-of-the-art high-dimensional multiobjective feature selection algorithms on 12 datasets. The results of HV and IGD show that MOFS-MST exhibits strong convergence and diversity. This can be attributed to the algorithm's ability to efficiently reduce the search space through feature grouping and quickly identify the regions containing valuable features. At the same time, the space transformation of multiple forms enables the population to search different directions in different spaces, which increases the diversity. The figure of Pareto front demonstrates that MOFS-MST can achieve higher-classification accuracy and diversity. Moreover, the results of average MCA and FN also show that the proposed algorithm prefers classification accuracy. It is proved that the proposed knowledge transfer strategy is effective, which can help the population to jump out of the local optimal and find better solutions. The running times also prove that the proposed algorithm is more efficient.

Although the proposed MOFS-MST has certain advantages, more work is needed to further explore its potential capabilities. First, the number of features selected is still large, so the potential value of selecting a small number of feature regions can be further explored. Second, grouping features according to redundancy, interactivity, and importance when initializing groups is an area worth exploring.

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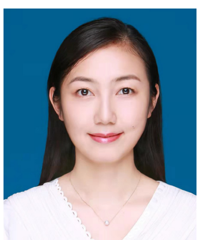

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